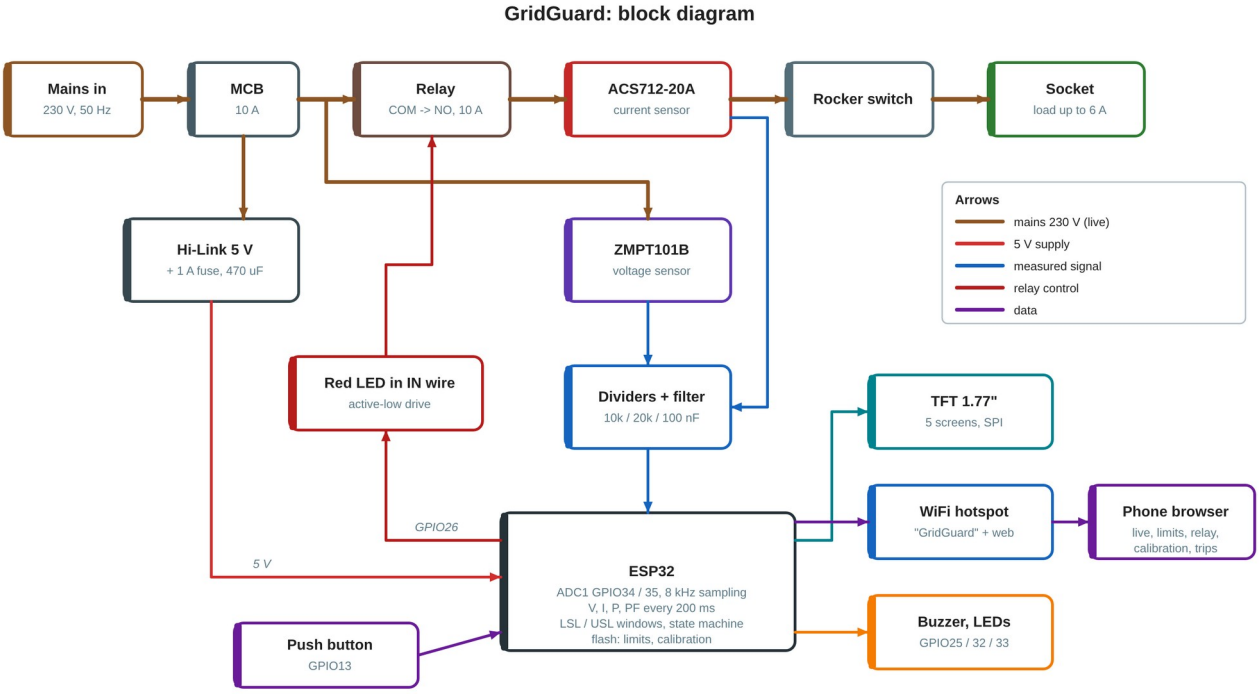
PROJECT DOCUMENT · BUILD, OPERATION AND REPORT

# GridGuard

ESP32 voltage, current and power protector with a relay cut-off




| | |
|---|---|
| Document | Build guide, standard operating procedure and project report |
| Firmware | GridGuard 0.4.0 (Arduino, ESP32) |
| Hardware | ESP32 DevKit, ZMPT101B, ACS712-20A, 5 V relay module, 1.77" ST7735S TFT, Hi-Link HLK-5M05 |
| Repository | github.com/shree-surya/GridGuard |
| Date / revision | October 2026, revision 1.0 |
| Prepared by | |

**Mains voltage**

This project switches 230 V AC. Build and test it only with supervision if you are new to mains wiring. Read Section 6.1 before you start. GridGuard is a learning project, not a certified protection device: the MCB stays as the real protection.

# Contents


| # | Section | What it covers |
|---|---|---|
| 1 | Introduction | aim, what GridGuard does and does not do |
| 2 | Specifications | ratings, limits, timings |
| 3 | How it works | block diagram, sensing, limits, states, relay drive |
| 4 | Components list | parts, quantities, approximate cost, tools |
| 5 | Wiring | pin map, low-voltage and mains diagrams, relay notes |
| 6 | Standard operating procedure | safety, assembly, checks, daily use, resets, maintenance |
| 7 | Flashing and first setup | Arduino IDE, libraries, upload, WiFi |
| 8 | Calibration and testing | voltage, current, zero; limit tests; acceptance checklist |
| 9 | Display and web page guide | TFT screens, button, dashboard |
| 10 | Troubleshooting | problems met while building and their fixes |
| 11 | Limitations and future work | |
| A–C | Appendices | web API, settings in flash, files |


# 1. Introduction

GridGuard sits between the wall socket and an appliance. It measures the mains voltage and the load current continuously and keeps a relay on the live wire closed only while voltage, current and power are all inside limits that you choose. If one of them stays outside its limit for 2 seconds, the relay opens, the buzzer sounds and the reason is logged. Everything is shown on a colour TFT and on a web page served over GridGuard's own WiFi hotspot, where you can also set the limits, switch the relay and calibrate the readings.

**Aim:** to protect appliances such as fridges, pumps and chargers from bad supply voltage and from overload or dry-running, with clear local and phone feedback, using low-cost parts.

### What GridGuard does not do

- It does not replace the MCB. Short circuits are cleared by the MCB; GridGuard reacts in seconds, not milliseconds.
- It is not certified and has no surge (lightning) protection.
- It is for single-phase loads up to 6 A (about 1.3 kW).

# 2. Specifications

**Table 1. GridGuard specifications**

| Item | Value |
|---|---|
| Supply and load | 230 V AC, 50 Hz, single phase; load up to 6 A |
| Switching | 5 V relay module, 10 A contacts, live wire, normally open (no power = load off) |
| Protection limits | lower (LSL) and upper (USL) limit for voltage, current and power, any values with LSL below USL; LSL 0 = not used |
| Default limits | voltage 190–250 V, current 0–6 A, power 0–1300 W |
| Trip delay | 2 s (a value must stay outside its window this long) |
| After a voltage trip | automatic: relay closes 10 s after the voltage is back inside its window |
| After a current or power trip | locked off until reset (button hold 5 s or web ON) |
| Start grace | current and power LSL checked from 5 s after the relay closes |
| Measurement | 1600 samples per channel in 200 ms (8 kHz); V, I, P, PF five times a second |
| Smallest readable load | about 0.2 A with WiFi on |
| Display | 1.77" ST7735S TFT, 128 x 160, SPI; five screens; Bai Jamjuree font |
| Controls | push button (screens, reset, relay off); web ON/OFF switch |
| Alarms | buzzer (3 beeps on a trip), optional green and red LEDs |
| Web | own WiFi "GridGuard" (password gridguard), http://192.168.4.1: live values, 60 s graph, limits table, relay switch, calibration, trip log |
| Kept after power-off | limits and calibration (ESP32 flash) |

# 3. How it works

Figure 1 (next page) shows the blocks. The live wire runs from the plug through the MCB, the relay contacts, the current sensor and the rocker switch to the socket. The voltage sensor and the 5 V supply are connected after the MCB but **before** the relay, so GridGuard keeps running and keeps measuring the supply after it has cut the load off.

## 3.1 Sensing and maths

- **Voltage:** ZMPT101B transformer module; **current:** ACS712-20A Hall sensor in the live wire (100 mV/A).
- Both outputs pass a 10 kΩ / 20 kΩ divider (5 V to 3.3 V) with a 100 nF filter into GPIO34 and GPIO35 (ADC1, works with WiFi on).
- Every 200 ms the ESP32 reads 1600 samples per channel (10 mains cycles), removes the DC midpoint and computes Vrms, Irms, real power P = mean(v × i) and power factor. Windows stretched by WiFi (outside 196–204 ms) are dropped.
- The sensor's idle noise (about 0.19 A) is removed in quadrature: $I = \sqrt{(I_{measured}^2 - I_{zero}^2)}$. Readings close to the noise count as 0 A.

## 3.2 Limits and states

Each quantity has a window from LSL to USL. Voltage is checked all the time. Current and power can only be seen while the relay is on, so they are checked only then, and their LSL (under-current, under-power, for example a pump running dry) only from 5 s after the relay closes, to let a motor start. Figure 4 in Section 5 shows the state machine; Table 3 lists the states.

**Table 2. Protection states**

| State | Relay | Meaning and what happens next |
|---|---|---|
| WAITING | off | power-up or after a voltage trip; counts 10 s while the voltage is inside its window, then closes the relay |
| ARMED | on | everything inside its window; watching |
| TRIPPING | on | a value is outside its window; counting 2 s. Back inside in time: ARMED again |
| TRIPPED | off | relay opened, 3 beeps, trip logged; shown for 3 s |
| LOCKED | off | after a current or power trip; stays off until reset (button hold 5 s or web ON) |
| OFF (manual) | off | turned off by hand (button hold 2 s or web OFF); stays off until turned on |

## 3.3 Relay drive

Most 5 V relay modules are **active-low**: IN at 0 V switches the relay on, IN left unconnected keeps it off. To switch them off through a wire, IN must be pulled almost to 5 V, which a 3.3 V ESP32 cannot do. GridGuard therefore pulls GPIO26 to 0 V for ON and releases the pin (input, nothing driven) for OFF, and a **red LED in series with IN** takes up about 1.8 V so that a released pin really means OFF. The LED doubles as a "relay on" light. Because the pin is released while the ESP32 boots, resets or is being flashed, the relay is always off then.

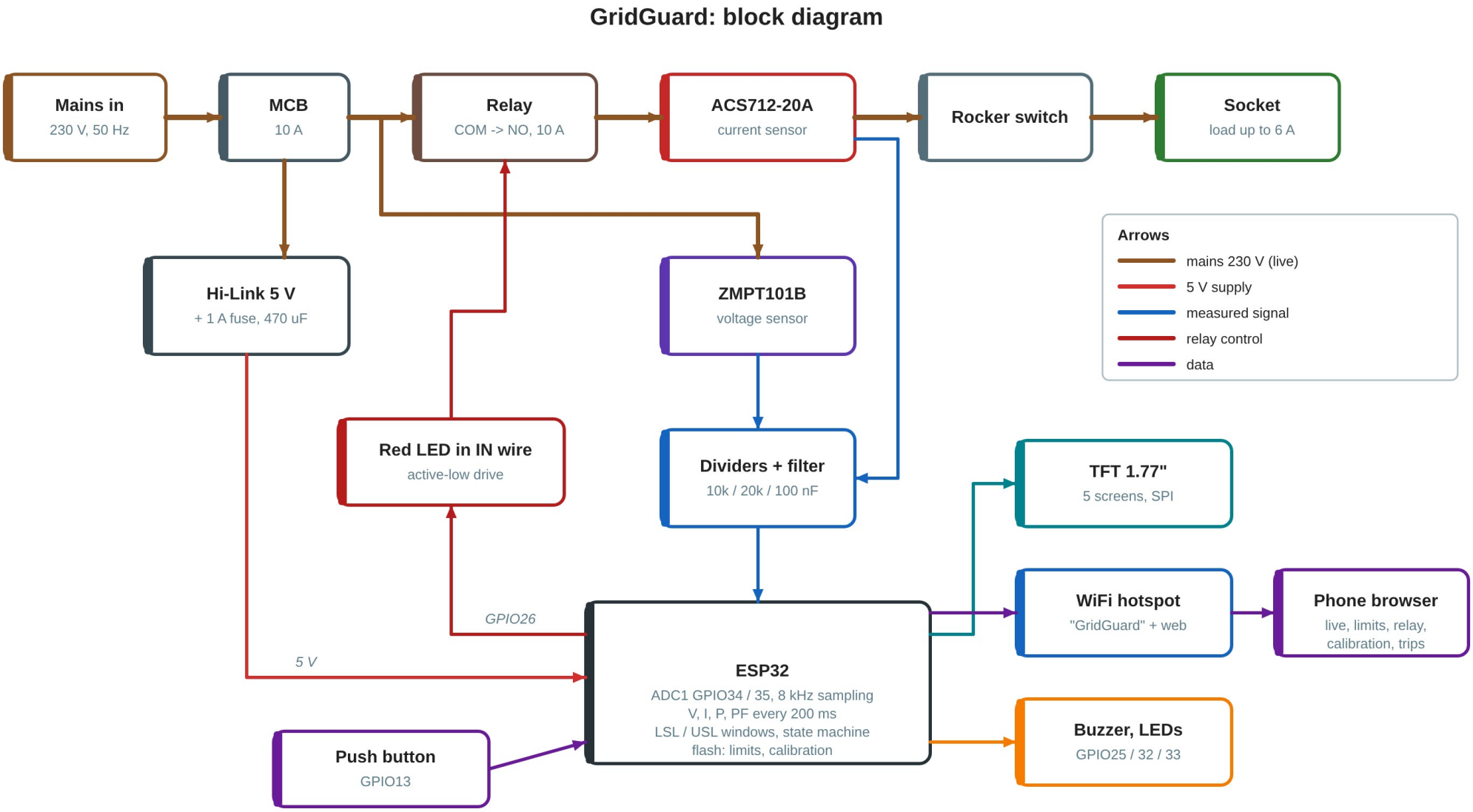



**Figure 1.** *GridGuard block diagram*

# 4. Components list

**Table 3. Bill of materials (one unit). Prices are approximate, in rupees.**

| # | Component | Specification | Qty | Purpose | Rs |
|---|---|---|---|---|---|
| 1 | ESP32 DevKit | 38-pin, ESP-WROOM-32, CP2102 USB | 1 | controller, WiFi, web server | 450 |
| 2 | Voltage sensor | ZMPT101B module, 5 V | 1 | measures mains voltage | 150 |
| 3 | Current sensor | ACS712-20A module, 5 V (100 mV/A) | 1 | measures load current | 120 |
| 4 | TFT display | 1.77" ST7735S, 128 x 160, SPI (8 pins incl. BLK) | 1 | local display | 350 |
| 5 | Relay module | 5 V, 1 channel, 10 A, relay marked SRD-05VDC-SL-C | 1 | switches the live wire | 80 |
| 6 | Red LED | 5 mm red, in series with relay IN | 1 | makes the relay switch off; on-indicator | 2 |
| 7 | Buzzer | 5 V active buzzer | 1 | trip alarm | 20 |
| 8 | Transistor + resistor | BC547 or 2N2222 NPN, 1 kΩ | 1 + 1 | drives the buzzer | 5 |
| 9 | Push button | 12 mm tactile or panel push button | 1 | screens, reset, relay off | 15 |
| 10 | Status LEDs (optional) | green + red 5 mm, 330 Ω each | 2 + 2 | ARMED / TRIP lamps | 10 |
| 11 | Power supply | Hi-Link HLK-5M05, 5 V 1 A | 1 | powers the electronics | 225 |
| 12 | Fuse + holder | 1 A slow-blow | 1 | protects the Hi-Link input | 40 |
| 13 | Resistors | 10 kΩ 1 % x2, 20 kΩ 1 % x2 | 4 | 5 V to 3.3 V dividers | 20 |
| 14 | Capacitors | 100 nF ceramic x3, 470 µF 16 V electrolytic x1 | 4 | filters; 5 V rail at VIN (required) | 20 |
| 15 | MCB | single pole, 10 A (6 A preferred) | 1 | short-circuit protection | owned |
| 16 | Socket, rocker switch | 6 A modular | 1 + 1 | load socket, manual ON/OFF | owned |
| 17 | Enclosure | modular plate + surface box | 1 | housing | owned |
| 18 | Plug and cord | 6 A 3-pin plug, 2 m 3-core cord | 1 | mains input | 50 |
| 19 | Wire and terminals | 1.5 sq mm load wire, hook-up wire, lugs or ferrules, terminal blocks, heat shrink | 1 set | connections | 200 |
| 20 | Board and hardware | perfboard, female headers, standoffs, cable gland, zip ties | 1 set | mounting | 200 |
| 21 | Test bulb | 60–100 W incandescent + holder | 1 | first mains test | 100 |
| | **Total (all parts new)** | | | | **about 2,100** |

**Check the relay before you buy or fit it**

The blue relay must read **SRD-05VDC-SL-C** (5 V coil). A 12 V relay (SRD-12VDC) lights its LED on 5 V but never clicks.

> A module with an **H/L trigger jumper** set to H also works, without the series LED: set RELAY_ACTIVE_LOW = false in the sketch and add 10 kΩ from GPIO26 to GND.

## Tools

- Digital multimeter and clamp meter
- Soldering iron, wire stripper, crimp tool, screwdrivers
- Socket tester
- Laptop with the Arduino IDE and a USB data cable
- Resistive test load (kettle or heater) and a 60–100 W bulb

# 5. Wiring

## 5.1 Pin map

Table 4. ESP32 connections

| ESP32 pin | Goes to | Note |
|---|---|---|
| 5V / VIN | 5 V rail (Hi-Link +Vo) | no series diode; 470 µF + 100 nF across VIN and GND |
| GND | GND rail | common ground |
| 3V3 | TFT VCC and BLK | 3.3 V output |
| GPIO34 | voltage divider node | ADC1, input only |
| GPIO35 | current divider node | ADC1, input only |
| GPIO23 | TFT SCK | SPI clock |
| GPIO19 | TFT SDA | SPI data (MOSI), not I2C |
| GPIO5 | TFT CS | |
| GPIO4 | TFT RST | |
| GPIO27 | TFT DC | |
| GPIO26 | relay IN through a red LED | LED long leg to IN, short leg to GPIO26; no resistor |
| GPIO25 | buzzer driver | 1 kΩ to the NPN base |
| GPIO13 | push button to GND | internal pull-up, no resistor |
| GPIO32 / GPIO33 | green / red LED | optional, 330 Ω each |

Table 5. Module connections

| Module | Pin | Connect to |
|---|---|---|
| ZMPT101B | VCC / GND / OUT | 5 V / GND / 10 kΩ to the GPIO34 node; L / N input after the MCB, before the relay |
| ACS712-20A | VCC / GND / OUT | 5 V / GND / 10 kΩ to the GPIO35 node; IP+ / IP− in the live wire after the relay |
| Relay module | VCC / GND / IN | 5 V / GND / red LED to GPIO26; COM to live from the MCB, NO to ACS712 IP+ |
| Buzzer | + / − | 5 V / NPN collector; NPN emitter to GND, base through 1 kΩ to GPIO25 |
| TFT | VCC, BLK / GND / CS, RST, DC, SDA, SCK | 3V3 / GND / GPIO5, 4, 27, 19, 23 |
| Hi-Link | AC L / AC N; +Vo / −Vo | live after the MCB via 1 A fuse / neutral; 5 V rail / GND |

**Notes**

Dividers: 10 kΩ from the sensor output to the node, 20 kΩ and 100 nF from the node to GND. Each node reads about 1.65 V with nothing flowing.

TFT pins are set at the top of the sketch (TFT_SDA = 19, TFT_SCK = 23). Change them there if your wires go elsewhere.

Pinouts of the NPN with the flat face towards you: BC547 is C B E, 2N2222 (TO-92) is E B C.

## 5.2 Diagrams

Figure 2 shows every low-voltage connection on the real 38-pin pin order. Figure 3 shows the 230 V path. Figure 4 shows the protection states. Full-size images are in the repository under docs/.

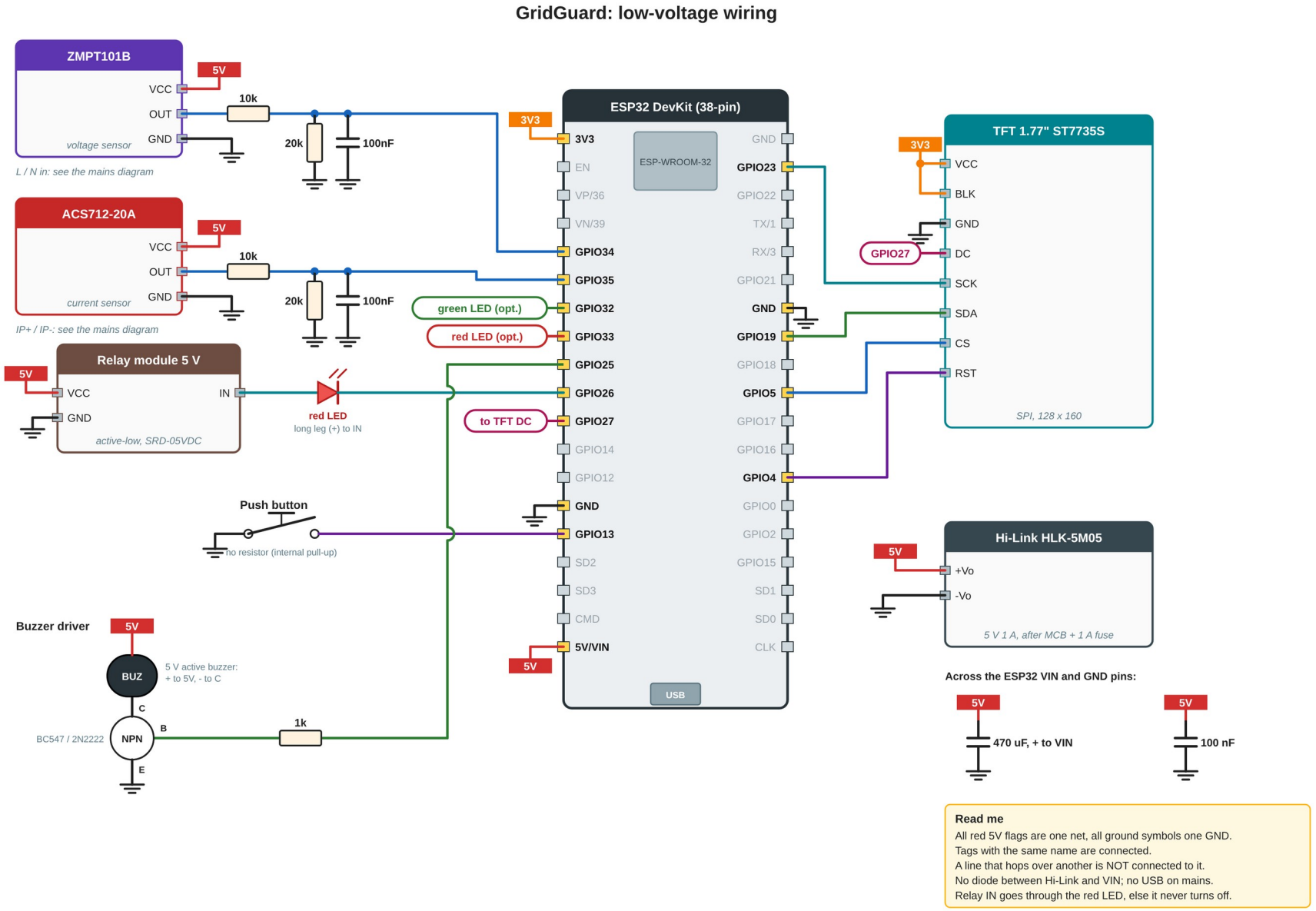
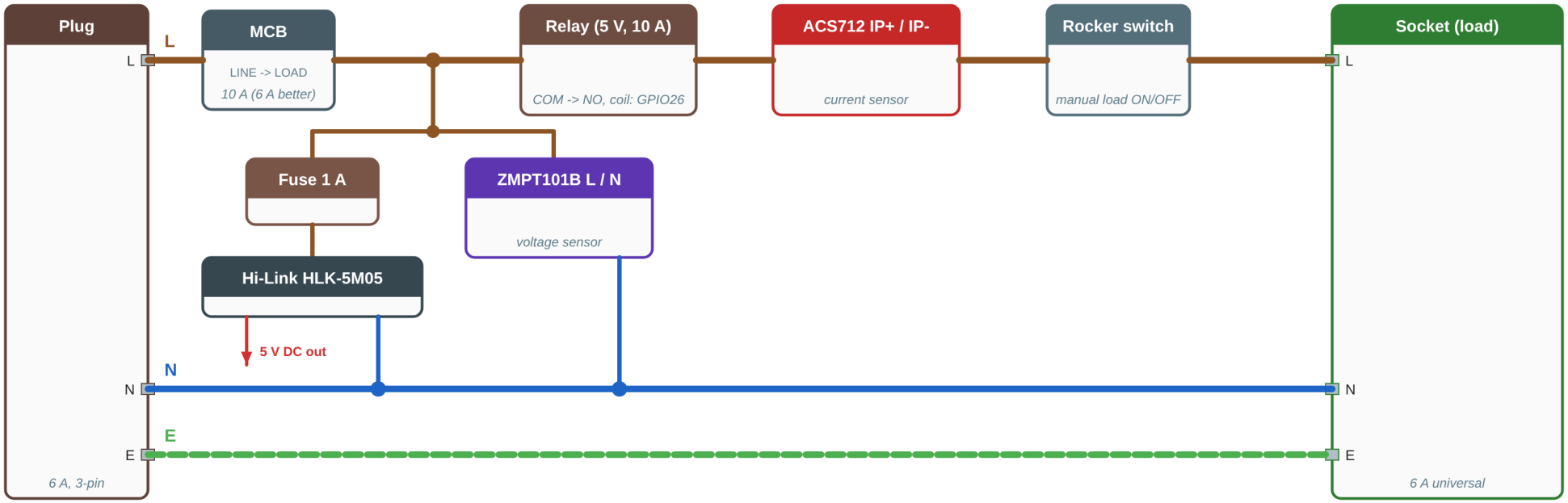



**Figure 2.** *GridGuard low-voltage wiring*

## GridGuard: mains wiring (230 V)




**Read me**

Live only through the MCB, relay (COM -> NO), ACS712 and rocker switch. Neutral and earth go straight through. NO contact: no coil power = socket off.
Hi-Link and ZMPT101B are taken after the MCB and BEFORE the relay, so GridGuard keeps running and measuring after a trip.
1.5 sq mm wire for the load path, crimped lugs or ferrules, every module inside the closed box. Test the wall socket for swapped L and N first.

**Figure 3.** *GridGuard mains wiring (230 V)*

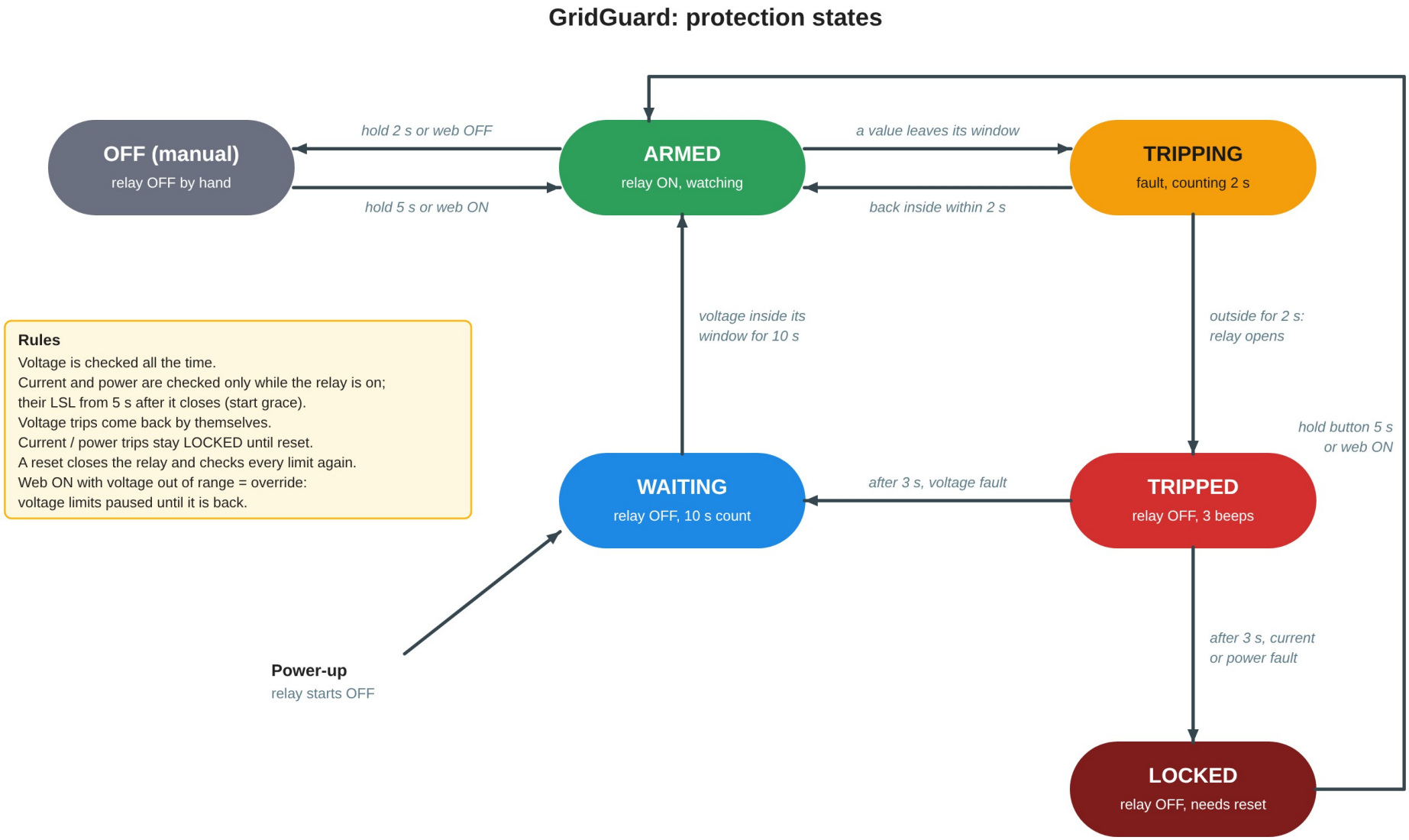



**Figure 4.** *GridGuard protection states*

# 6. Standard operating procedure

## 6.1 Safety rules

- Work on the box only when it is **unplugged**. The MCB off is not enough.
- **Never connect USB while the box is on mains.** The ESP32 is not isolated from the Hi-Link output.
- Relay on the live wire only, contacts COM and **NO** (normally open). Neutral and earth go straight through.
- Keep the MCB: GridGuard is not a short-circuit protector.
- No bare modules outside a closed box; cover terminals; 1.5 sq mm wire on the load path.
- Test the wall socket for swapped live and neutral before connecting.
- First mains power-up through a 60–100 W bulb in series (Section 6.4).

## 6.2 Assembly

1. Mount the modules in the plate and box; fit a cable gland for the cord.
2. Solder the ESP32 headers, the two dividers and the 5 V / GND rails on the perfboard; fit 470 µF + 100 nF across VIN and GND.
3. Wire the TFT (Table 4), the push button (GPIO13 to GND), the buzzer driver (GPIO25, 1 kΩ, NPN) and the optional LEDs.
4. Wire the relay module: VCC to 5 V, GND to GND, IN through the red LED to GPIO26 (long leg to IN).
5. Wire the Hi-Link output straight to the 5 V and GND rails (no diode).
6. Mains: plug L to MCB LINE; MCB LOAD to relay COM; relay NO to ACS712 IP+; IP− to the rocker switch; switch to socket L. N and E straight to the socket.
7. From MCB LOAD (before the relay): through the 1 A fuse to Hi-Link AC L, and to ZMPT101B L. Neutral to Hi-Link AC N and ZMPT101B N.
8. Keep mains and low-voltage wiring apart; tie them down; fit terminal covers.

## 6.3 Checks before the first power-up

| # | Check | Expected | Result |
|---|---|---|---|
| 1 | 5 V rail to GND, 3V3 to GND (unplugged) | not a short | ☐ pass ☐ fail |
| 2 | Any mains point to ESP32 GND | open (OL) | ☐ pass ☐ fail |
| 3 | Relay COM to NO, relay unpowered | open | ☐ pass ☐ fail |
| 4 | Plug L to socket L, relay unpowered | open (the relay is normally open) | ☐ pass ☐ fail |
| 5 | Plug N to socket N, plug E to socket E | continuity | ☐ pass ☐ fail |
| 6 | Red LED orientation at relay IN | long leg to IN, short leg to GPIO26 | ☐ pass ☐ fail |
| 7 | Relay marking | SRD-05VDC-SL-C (5 V) | ☐ pass ☐ fail |

## 6.4 First power-up

1. **USB only, no mains, nothing on the relay contacts.** Flash GridGuard_hw_test.ino first (Section 7): the screen works, each button press toggles the relay with a click and a beep, both sensor midpoints read about 1650 mV.
2. Press the ESP32's EN (reset) button: the relay must go off. Check COM–NO with the meter: beeps only when ON.

3. Flash GridGuard.ino. With no mains the relay stays off and the screen shows BAD SUPPLY, because 0 V is outside the voltage window.
4. Unplug USB. Plug into mains through a **60–100 W bulb in series**. The bulb glows dimly or flashes once, then stays dark. Bright and steady means a short.
5. Remove the bulb, plug in directly, MCB on. After 10 s with a good voltage the relay clicks on (ARMED). Then calibrate (Section 8).

## 6.5 Daily operation

1. Check that the box is closed and the cord is undamaged.
2. Plug GridGuard into the wall socket and switch the MCB on. The screen shows STARTING 10 s; the relay closes by itself if the voltage is inside its window.
3. Plug the appliance into the front socket and switch the rocker on.
4. Watch the Home screen: green ARMED means protected. Short-press the button to see the graph, trips, limits and WiFi details.
5. Optional: join WiFi **GridGuard** (password **gridguard**) and open **http://192.168.4.1** to watch, set limits or switch the relay.
6. To finish: rocker off, unplug the appliance, MCB off, unplug GridGuard.

## 6.6 Setting the limits

1. Open the web page, section **Limits**. For each of voltage, current and power, type LSL (lower) and USL (upper). Use 0 as LSL to switch a lower limit off.
2. Press **Save limits**. They are checked (LSL must be below USL), saved in flash and shown on the Limits screen.
3. Starting points: fridge 190–250 V, 0–6 A; water pump 190–250 V, 0.5–6 A (LSL catches dry running); charger 190–250 V, 0–3 A.

## 6.7 After a trip, and resets

| Trip reason | What to do |
|---|---|
| Over- or under-voltage | nothing: the relay closes by itself 10 s after the voltage is back inside its window |
| Over- or under-current, over- or under-power | find the cause (overload, jammed motor, dry pump). Then reset: hold the button 5 s (the screen counts RESETTING 5 4 3 2 1; let go to cancel) or tap ON on the web page. The relay closes and every limit is checked again. |
| Turned off by hand | hold the button 5 s, or tap ON on the web page |

> **Web ON while the voltage is out of range**
>
> Tapping ON on the web page while the voltage is outside its window starts an **override**: the relay closes and the voltage limits are ignored until the voltage comes back inside them. Current and power protection stay on. Use it only when you know the appliance can take that voltage. The button reset never overrides: it checks all limits straight away.

## 6.8 Maintenance

- Monthly (unplugged): terminal screws tight, no discoloured wires, relay contacts not buzzing.
- Every six months: compare voltage and current with meters; recalibrate if they differ by more than 2 %.

- Test the protection now and then: set the voltage USL just below the mains voltage and check that it trips, then set it back.

# 7. Flashing and first setup

1. Arduino IDE with the ESP32 board package; board **ESP32 Dev Module**.
2. **Tools > Partition Scheme > Huge APP (3MB No OTA/1MB SPIFFS)**.
3. Libraries: **Adafruit GFX Library** and **Adafruit ST7735 and ST7789 Library** (Library Manager); **ESPAsyncWebServer** and **AsyncTCP** by ESP32Async (GitHub > Code > Download ZIP > Sketch > Include Library > Add .ZIP Library).
4. Folder **GridGuard** with **GridGuard.ino**, **BaiJamjuree.h** and **web_index.h** (firmware/GridGuard in the repository). Open the .ino.
5. Check the settings at the top of the sketch: `TFT_SDA = 19`, `TFT_SCK = 23`, `RELAY_ACTIVE_LOW = true`.
6. USB connected, **mains unplugged**; Upload. Serial Monitor at 115200, line ending "Newline".
7. Phone: WiFi **GridGuard** / **gridguard**, open **http://192.168.4.1**.

**Table 6. Sketches in the repository**

| Sketch | Use |
|---|---|
| GridGuard_display_test | screens only, simulated readings, no sensors (checks the TFT wiring) |
| GridGuard_hw_test | one screen with sensor midpoints; button toggles the relay and beeps (checks all wiring) |
| GridGuard | the full firmware |

# 8. Calibration and testing

Open the **Calibration** section of the web page (tap to expand). Values are saved in flash immediately. The same works from the Serial Monitor: `V 231.5`, `I 4.20`, `O -2.5`, `Z`, `F`, `D` (defaults), `?`.

## 8.1 Calibration steps

1. **Voltage:** measure the socket with a multimeter, type it under "Multimeter reads" and press **Match**. This sets the gain; the offset (−30 to +30 V) can also be typed.
2. **Current:** run a kettle or heater (at least 0.5 A) with the relay on, type the clamp meter reading under "Clamp meter reads" and press **Match**.
3. **Zero offset:** unplug the load and press **Measure zero** (about 4 s). It refuses if a load still draws current. An empty socket then reads 0.00 A.
4. **Direction:** press **Flip direction** if power reads wrong on a known load.
5. **Defaults** restores the factory calibration; the limits are not changed.

**Table 7. Calibration values (defaults)**

| Value | Default | Range / meaning |
|---|---|---|
| Voltage gain | 0.946 | 0.5 to 1.5; mains volts per mV at the pin |
| Voltage offset | 0 V | −30 to +30 V added after the gain |
| Current trim | 1.00 | 0.5 to 2.0; clamp meter A / shown A |

| Value | Default | Range / meaning |
|---|---|---|
| Zero offset | 0.195 A | 0 to 1 A; idle noise removed in quadrature |
| Direction | −1 | +1 or −1 |

## 8.2 Protection tests

| # | Check | Expected | Result |
|---|---|---|---|
| 1 | Power-up with good mains | STARTING 10 s, then ARMED, relay clicks on | ☐ pass ☐ fail |
| 2 | Voltage USL set just below the mains voltage | TRIP IN 2 s, TRIPPED, 3 beeps, relay off; WAITING; back on 10 s after USL is restored | ☐ pass ☐ fail |
| 3 | Current USL 0.5 A with a kettle on | trips, then LOCKED (HOLD 5s RESET) | ☐ pass ☐ fail |
| 4 | Hold the button 3 s, release | RESETTING countdown, then "Reset cancelled", stays LOCKED | ☐ pass ☐ fail |
| 5 | Hold the button 5 s (USL back to 6 A) | relay on, ARMED | ☐ pass ☐ fail |
| 6 | Current LSL 0.5 A, nothing plugged in | after 5 s grace and 2 s delay: under-current trip, LOCKED | ☐ pass ☐ fail |
| 7 | Web OFF, then web ON | relay off (OFF manual), then on (ARMED) | ☐ pass ☐ fail |
| 8 | Button hold 2 s with relay on | relay off (OFF manual) | ☐ pass ☐ fail |
| 9 | ESP32 EN (reset) button | relay off while the ESP32 restarts | ☐ pass ☐ fail |
| 10 | Power-cycle | limits and calibration kept; trip log cleared | ☐ pass ☐ fail |
| 11 | Multimeter / clamp meter vs GridGuard | within 1 % / 2 % after calibration | ☐ pass ☐ fail |

# 9. Display and web page guide

## 9.1 TFT screens and button

A short press of the button shows the next screen; otherwise the screen stays where it is. A trip always brings Home back. Holding the button: with the relay on, 2 s turns it off; with the relay off, 5 s resets (RESETTING 5 4 3 2 1 on the screen, a tick each second).

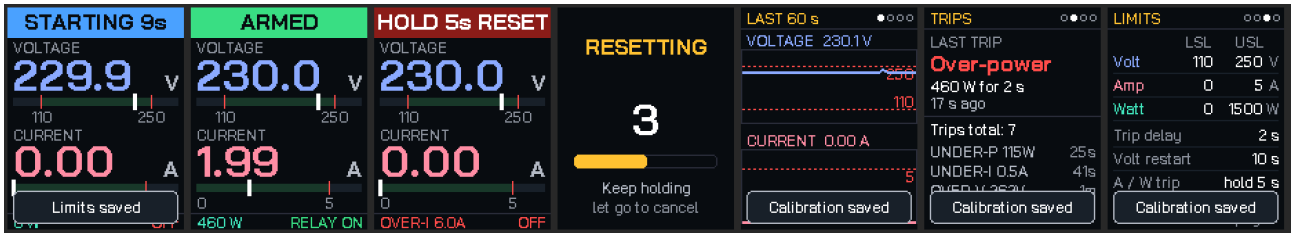
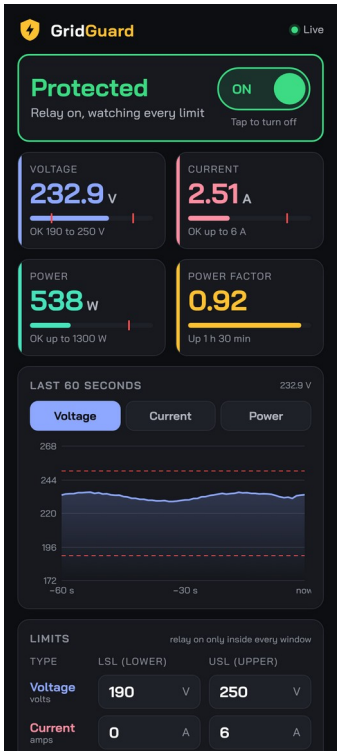


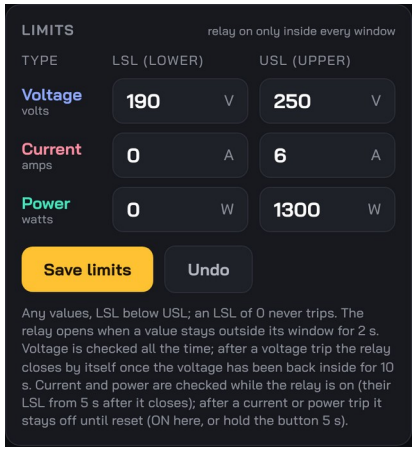

**Figure 5.** *TFT screens: STARTING, ARMED, LOCKED, RESETTING countdown, Graph, Trips, Limits*

| Screen | Shows |
|---|---|
| Home | status bar (ARMED green, TRIP IN amber, TRIPPED red, WAITING / STARTING blue, LOCKED dark red, OFF grey), voltage and current with bars showing the LSL–USL window, power and relay state |
| Graph | last 60 s of voltage and current with the limits as dashed lines |
| Trips | last trip with reason, value and time, total since power-up, earlier trips |
| Limits | LSL and USL of voltage, current and power; trip and restart delays |
| Connect | WiFi name, password, address, phones connected, firmware version, button help |

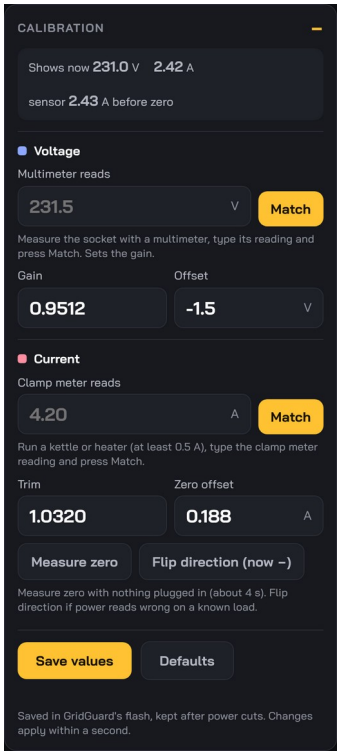
## 9.2 Web dashboard




*Status, relay switch, live values*




*Limits table (LSL / USL)*




*Calibration*

**Figure 6.** *GridGuard web page*

| Part | What you can do |
|---|---|
| Status card | state in words with the reason and countdown; the ON/OFF switch turns the relay on or off (OFF asks to confirm) |
| Live tiles | voltage, current, power with their windows marked, power factor and uptime |
| Last 60 seconds | graph of voltage, current or power with the limits |
| Limits | LSL and USL for each quantity; Save limits / Undo |
| Calibration | match to meters, gain, offset, trim, zero offset, measure zero, flip, defaults |
| Trip log | last 10 trips with reason, value and time ago |

# 10. Troubleshooting

| Problem | Likely cause | Fix |
|---|---|---|
| TFT stays white | SDA / SCK swapped or on other pins than the sketch; RST / DC / CS loose | check wiring against TFT_SDA / TFT_SCK at the top of the sketch; run GridGuard_display_test; BLK to 3V3 |
| Picture shifted, mirrored or red / blue swapped | different ST7735 panel variant | try INITR_GREENTAB or INITR_REDTAB in initR(), see the notes in the display test |
| Relay always ON, even when off | active-low module: a released 3.3 V pin still looks low | put a red LED in series with IN (long leg to IN); RELAY_ACTIVE_LOW = true |
| Relay ticks or buzzes | IN driven to 3.3 V, half on | same fix: series LED and the current firmware |
| Relay LED lights but no click | 12 V relay (SRD-12VDC) on 5 V, or JD-VCC not powered | use a 5 V relay; on JD-VCC boards power JD-VCC from 5 V or refit the jumper |
| Works on USB, resets on the Hi-Link | series diode dropped VIN to about 3.6 V | no diode; 470 µF at VIN |
| Compile error "declared void" | old sketch with types below the first function | use the current sketches from the repository |
| "Sketch too big" | default partition | Huge APP (3MB No OTA/1MB SPIFFS) |
| Relay never closes after power-up (BAD SUPPLY) | voltage outside its LSL–USL window, or voltage not calibrated | check the limits and calibrate the voltage |
| Keeps showing HOLD 5s RESET | a current or power trip; locked on purpose | fix the cause, then hold 5 s or tap ON |
| Under-current trips right after reset | current LSL above the load's real current | lower or zero the current LSL; the start grace is 5 s |
| Current match refused | less than 0.5 A flowing | use a kettle or heater with the relay on |
| Measure zero refused | a load is still connected | unplug the load and try again |
| Small loads read 0 A | below the noise floor (about 0.2 A) | expected with the 20 A sensor |

# 11. Limitations and future work

- Reaction time is seconds (trip delay 2 s); the MCB handles short circuits.
- No surge protection; an MOV ahead of the MCB is a possible addition.

- The trip log lives in RAM and is cleared on power-up; a dated log in flash (time from the phone) is planned.
- Firmware 0.4.0 is checked in a PC simulation (36 automatic checks) and in a browser; the display, button and relay drive have been confirmed on the bench, calibration and the full trip tests on hardware are still to be done.
- Possible additions: PIN for settings, adjustable trip and restart delays, presets per appliance, OTA updates.

# Appendix A. Web API

| Route | Purpose |
|---|---|
| GET / | the web page (gzip) |
| GET /data | v, i, p, pf, st (state), relay, left (countdown), why, vl, vu, il, iu, pl, pu, td, rd, gs, trips, up, fw, ovr |
| GET /history | {dt, v[], i[], p[]}, last 60 s |
| GET /trips | {total, list[{why, val, unit, ago}]} |
| POST /limits?vl=&vu=&il=&iu=&pl=&pu= | set the limits (LSL below USL) |
| POST /relay?on=1\|0 | relay on (reset / override) or off |
| GET /cal | calibration values, live readings, zero state |
| POST /cal?... | vref=, iref=, vgain=, voff=, itrim=, inoise=, isign= or defaults=1 |
| POST /zero | measure the zero offset (load unplugged) |

# Appendix B. What is kept in flash

| Data | Saved |
|---|---|
| Limits (LSL / USL of V, I, P) | when you press Save limits |
| Calibration (gain, offset, trim, zero, direction) | when changed |
| Trip log, graph history | not saved (RAM only) |

# Appendix C. Repository files

| Path | Contents |
|---|---|
| firmware/GridGuard/ | GridGuard.ino, BaiJamjuree.h, web_index.h |
| firmware/GridGuard_hw_test/ | wiring test sketch |
| firmware/GridGuard_display_test/ | screen test sketch |
| web/index.html | the web page (self-contained) |
| tools/ | embed_web.py (page to web_index.h), ttf2gfx.py (font converter) |
| docs/ | this document, diagrams, wiring notes, spec |